

→ Bank.

1. I went to Bank to deposit money.

2. I went to Bank to sit & relax.

⇒ Numbers.

Semantic Understanding

→ Identifying what the text means, not just which words are used.

Ex: purchase a vehicle

ⓧ
buy an automobile

embeddings.

↳ Numerical representation of text that captures its semantic meaning.

↳ list of Numbers.

↳ Vector.

Dog $\rightarrow$ 79 $\rightarrow$ [10, 20] $\Rightarrow$ Vector Embedding.

Pen $\rightarrow$ 65

Apple $\rightarrow$ 47

$\rightarrow$ [-----]
1536.

gpt small model $\rightarrow$ 1536 dimensions

gpt large model $\rightarrow$ 3072 dimensions

Text to Vector Conversion.

Input text is given.

Ex: Best restaurants in Bangalore.

1) Break the text into smaller units.

$\rightarrow$ Tokenization.

Tokens:

$\rightarrow$ Best

$\rightarrow$ restaurants

$\rightarrow$ in

$\rightarrow$ Bangalore

II) Trained model converts these tokens into vectors.

↳ list of numbers.

→ Best : $[0.12 \quad -0.44 \quad 0.5 \quad 0.71 \quad 0.01 \quad \dots \dots \dots]$ ↖ 1536.
→ restaurants $[0.9 \quad 0.04 \quad -0.9 \quad -0.17 \quad 0.98 \quad \dots \dots \dots]$ ↖ 1536.
→ in
→ Bangalore

dog : $[0.4 \quad 0.17 \quad 0.02 \quad -0.65]$

puppy : $[0.45 \quad 0.21 \quad 0.05 \quad -0.57]$

Cat : $[0.11 \quad 0.72 \quad -0.31 \quad 0.07]$

Snake: $[0.91 \quad -0.99 \quad 0.12 \quad 0.15]$

apple : $[0.7 \quad 0.41 \quad -0.52 \quad 0.92]$

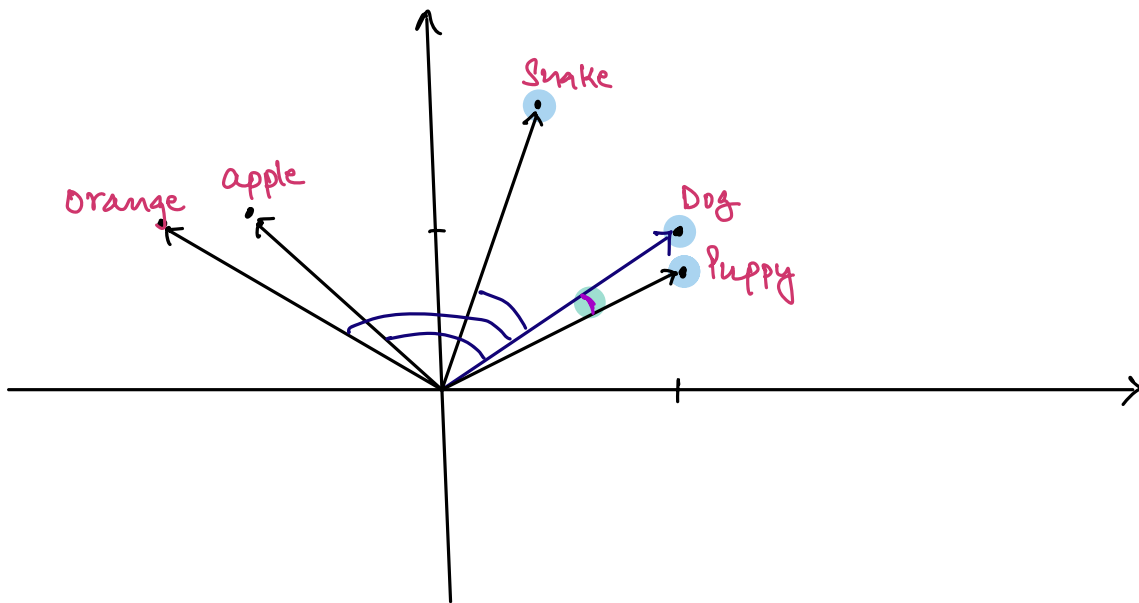
Orange: $[0.65 \quad 0.38 \quad -0.49 \quad 0.01]$

Words | Text with similar meaning are placed closer to each other in Vector Space.

Similar meaning $\Rightarrow$ Close Vectors.

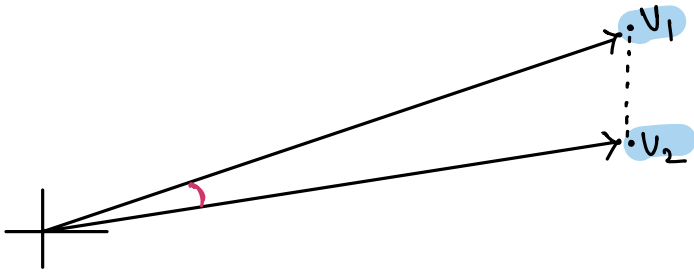
Different meaning $\Rightarrow$ far Vectors.

$[x, y] \rightarrow$ 2D Vector space



Similarity Measurement.

Based on how close two vectors are.



Similarity b/w 2 vectors can be measured :

→ distance

→ angle.

Keyword Search (vs) Semantic Search

↓
Exact word match

↳
meaning / context /
Similarity

Semantic Search Workflow.

- I) Store documents.
- II) Convert these documents into Vector Embedding.
- III) User asks a query.
- IV) Convert the query into Vector Embedding.
- V) Compare the query Vector Embedding with the Vector Embeddings of the documents.
- VI) Retrieve the document with the nearest Vector Embedding.

Embeddings in Agentic Systems.

Benefits / Applications of embeddings.

→ Semantic Search.

→ Recommendation System.

→ Clustering or Grouping.

→ Document retrieval.